

COURSE OBJECTIVES:

1. To know the details of data warehouse Architecture
2. To understand the OLAP Technology
3. To understand the partitioning strategy
4. To differentiate various schema
5. To understand the roles of process manager & system manager

UNIT I INTRODUCTION TO DATA WAREHOUSE

Data warehouse Introduction - Data warehouse components- operational database Vs data warehouse – Data warehouse Architecture – Three-tier Data Warehouse Architecture -Autonomous Data Warehouse- Autonomous Data Warehouse Vs Snowflake - Modern Data Warehouse.

UNIT II ETL AND OLAP TECHNOLOGY

What is ETL – ETL Vs ELT – Types of Data warehouses - Data warehouse Design and Modeling -Delivery Process - Online Analytical Processing (OLAP) - Characteristics of OLAP – Online Transaction Processing (OLTP) Vs OLAP - OLAP operations- Types of OLAP- ROLAP Vs MOLAP Vs HOLAP.

UNIT III META DATA, DATA MART AND PARTITION STRATEGY

Meta Data – Categories of Metadata – Role of Metadata – Metadata Repository – Challenges for Meta Management - Data Mart – Need of Data Mart- Cost Effective Data Mart- Designing Data Marts- Cost of Data Marts- Partitioning Strategy – Vertical partition – Normalization – Row Splitting– Horizontal Partition.

UNIT IV DIMENSIONAL MODELING AND SCHEMA

Dimensional Modeling- Multi-Dimensional Data Modeling – Data Cube- Star Schema- Snowflakeschema- Star Vs Snowflake schema- Fact constellation Schema- Schema Definition – Process Architecture- Types of Data Base Parallelism – Datawarehouse Tools.

UNIT V SYSTEM & PROCESS MANAGERS

Data Warehousing System Managers: System Configuration Manager- System SchedulingManager - System Event Manager - System Database Manager - System Backup Recovery Manager - Data Warehousing Process Managers: Load Manager – Warehouse Manager- Query Manager – Tuning – Testing.